

Calculate the Jacobian matrix of the following function:

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}^3 : f(x_1, x_2) = \begin{pmatrix} x_1^2 x_2 \\ \cos(e^{x_1 + x_2^2}) \\ x_2 \end{pmatrix}.$$